Semester VI **B. Tech CSE (AI)**

BCSAI 607: INTERNET OF THINGS

2L + 0T + 0P + 2C **MM 100**

Unit 1: Introduction to IoT

Defining IoT, Characteristics, Physical Design, Logical Design, Functional Blocks, Communication Models and APIs, Computer Networks, Internet of Everything (IOE), Distributed Computing, Industrial Automation, Understanding IT and OT Convergence, Evolution of IIoT, Machine to Machine Communication, Difference between IoT and M2M, Software Define Network.

Unit 2: Concept of Data, Information, Knowledge and Wisdom

Knowledge Discovery Process, DIKW Pyramid and Relevance with IoT, Microcontrollers, Cost, Performance and Power Consumption, Commercial Microcontroller Based Development Boards, Selection Criteria and Trade-Offs.

Unit 3: Network and Communication Aspects

Wireless Medium Access Issues, Mac Protocol Survey, Survey-Routing Protocols, Sensor Deployment Basics & Node Discovery, Data, Aggregation & Dissemination. Sensor Node Architecture, WSN/M2M Communication Technologies, Bluetooth, Zigbee, Wifi. Cellular Communication and LpWAN (Lora and LoraWAN Technologies), Topologies, Applications.

Unit 4: Design and Development of IoT Systems

IoT Reference Architectures, Standardization Initiatives and Interoperability Issues. IoT Design Considerations, Architecture (Devices, Networks and Cloud). Network, Communication Technologies and Protocols, Smart Asset Management: Connectivity, Visibility, Analytics and Alerts.

Unit 5: Domain Specific Applications of IoT and its Challenges

Home Automation, Industry Applications, Surveillance Applications, Other IoT Applications, Design Issues and Challenges in IoT, Security, Development, Deployment, Usage. Security Standards, Vulnerabilities, Attack Surfaces, Hardware and Software Solutions, Open Source Initiatives.

Text/ Reference Books:

1. Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle. (2014). From Machine-To-Machine to the Internet of Things: Introduction to a New Age of Intelligence. Academic Press.
2. Francis Dacosta. (2013). Rethinking the Internet of Things: A Scalable Approach to Connecting Everything. Apress.
3. Vijay Madisetti and ArshdeepBahga. (2014). Internet of Things: A Hands-On-Approach. Orient Blackswan.
4. Adrian Mcewen, HakinCassimally. (2015). Designing the Internet of Things. Wiley.
5. Peter Waher. (2018). Mastering Internet of Things: Design and create your own IoT applications using Raspberry Pi 3. Packt
6. Rolf H. Weber, Romana Weber. (2010). Internet of Things: Legal Perspectives. Springer.